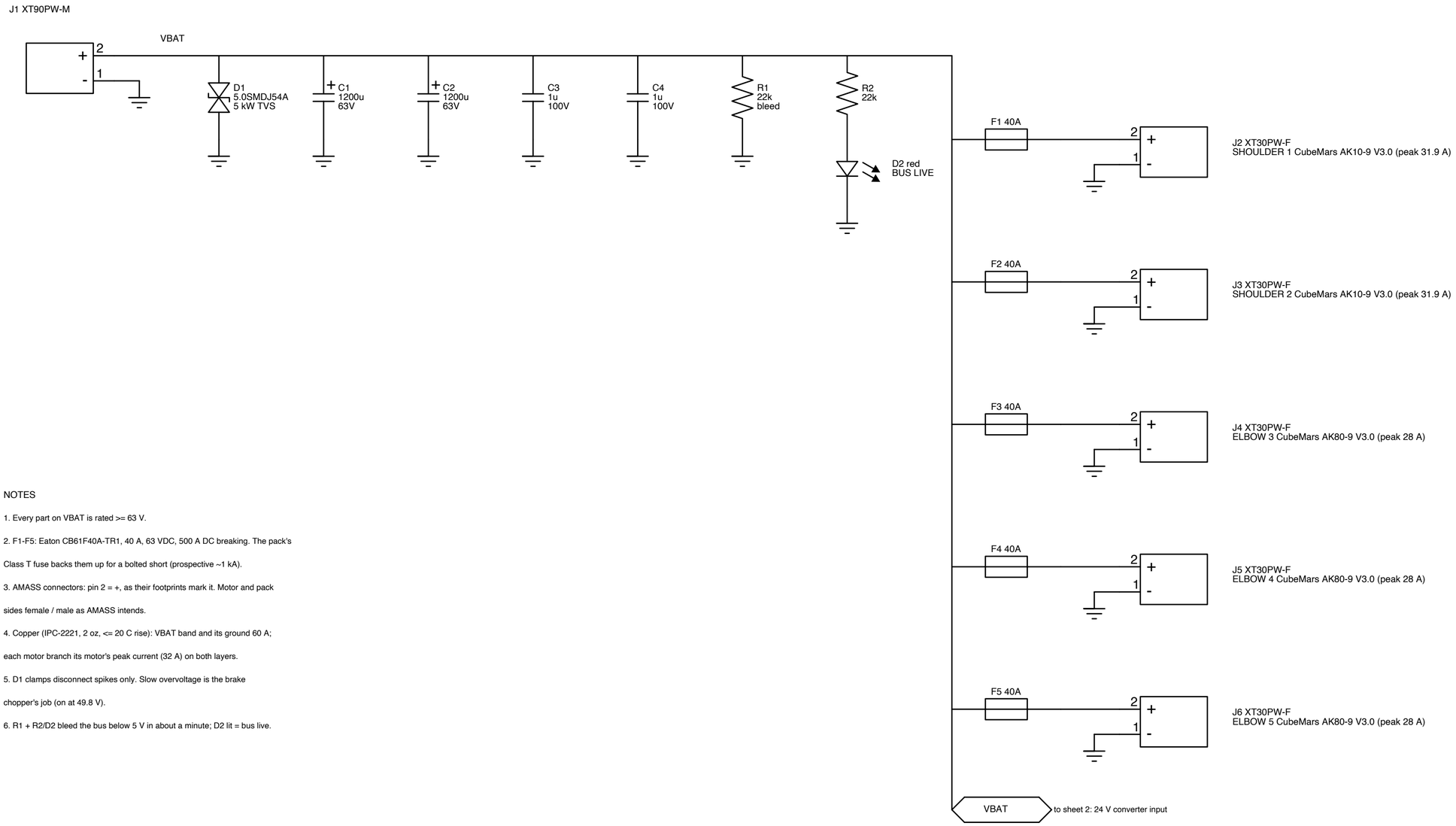
PACK INPUT - 12S Li-ion, 36.0-49.0 V (BMS OVP 50.4 V, drivers 52 V abs. max)

MOTOR FEEDS - 49 V, one 40 A fuse per motor



NOTES

1. Every part on VBAT is rated >= 63 V.

2. F1-F5: Eaton CB61F40A-TR1, 40 A, 63 VDC, 500 A DC breaking. The pack's Class T fuse backs them up for a bolted short (prospective ~1 kA).

3. AMASS connectors: pin 2 = +, as their footprints mark it. Motor and pack sides female / male as AMASS intends.

4. Copper (IPC-2221, 2 oz, <= 20 C rise): VBAT band and its ground 60 A; each motor branch its motor's peak current (32 A) on both layers.

5. D1 clamps disconnect spikes only. Slow overvoltage is the brake chopper's job (on at 49.8 V).

6. R1 + R2/D2 bleed the bus below 5 V in about a minute; D2 lit = bus live.

POWER DISTRIBUTION

PIONEER-ARM-PDB (cad_bench/projects/arm_pdb.py)

drawn by the harness; checked against the layout netlist

SHEET 1 / 2	REV A
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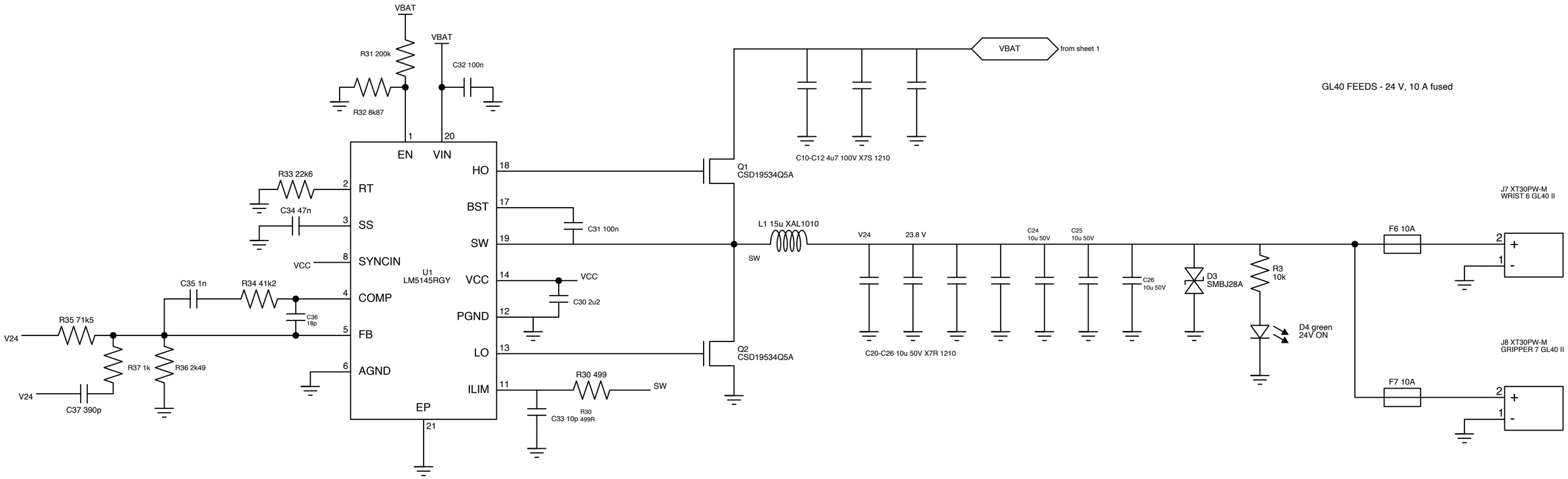
24 V BUCK - TI LM5145 (SNVSAI4B, application circuit 3: 48 V -> 24 V, 440 kHz, 5 A cont.)



NOTES

1. Values exactly as TI's application circuit 3 (compensation is TI's); RFB1 71k5 (E96) sets 23.8 V.

2. SYNCIN tied to VCC: forced PWM, so GL40 regen flows back to VBAT instead of raising V24.

3. 5 A continuous; the valley limit (R30 on Q2's RDS(on)) lets ~8-9 A out for peaks.

4. UVLO: R31 / R32 start the converter at 28.3 V, stop it at 26.3 V.

5. J7 / J8 are XT30 male: a GL40 lead cannot reach a 49 V socket, nor an AK lead a 24 V one.

24 V CONVERTER + GL40 FEEDS	
PIONEER-ARM-PDB (cad_bench/projects/arm_pdb.py)	
TI LM5145 application circuit 3, unchanged	
SHEET 2 / 2	REV A
DATE 2026-11-05	SIZE A3